radxa

Radxa ROCK 2F schematic

V1.01

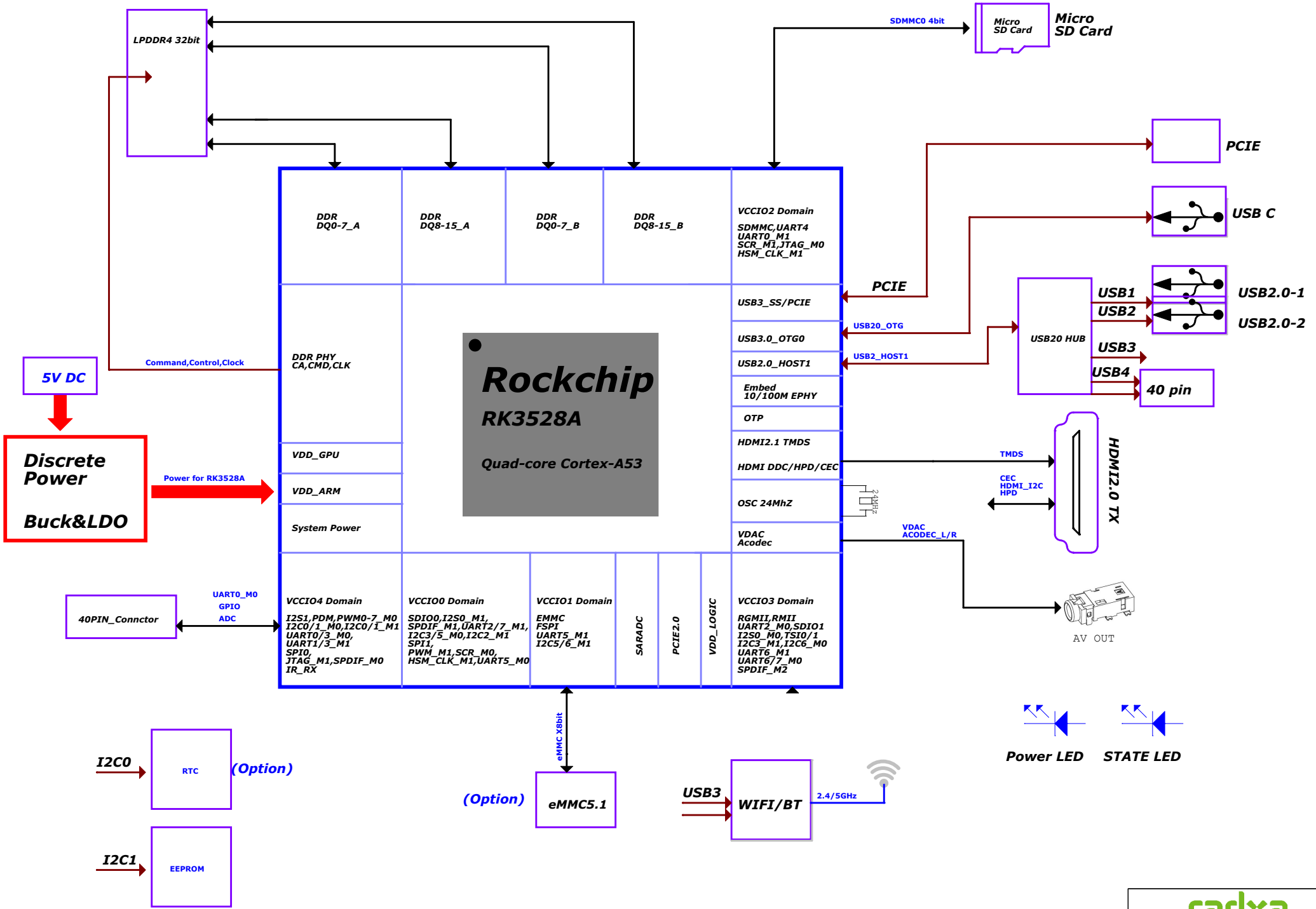
Index

Page 1	01.Cover
Page 2	02.Index
Page 3	03.Block
Page 4	04.RK3528A Power/GND
Page 5	05.RK3528A OSC/PLL/PMUIO
Page 6	06.RK3528A DDR Controller
Page 7	07.RK3528A FLASH/SD Controller
Page 8	08.RK3528A USB/PCIE Controller
Page 9	09.RK3528A SARADC/OTP
Page 10	10.RK3528A HDMI Interface
Page 11	11.RK3528A AV OUT
Page 12	12.RK3528A Embed FEPHY
Page 13	13.RK3528A_1.8V/3.3V GPIO
Page 14	14.Power-DiscretePower
Page 15	15.DRAM-LPDDR4_1X32bit_200P
Page 16	16.Flash-eMMC
Page 17	17.HDMI TX
Page 18	18.USB_HUB
Page 19	19.RTC_EEPROM
Page 20	20.WIFI/BT
Page 21	21.Micro_SD
Page 22	22.VO-AV Interface
Page 23	23.40PIN_Connctor_PCIE_CONN.
Page 24	24.USB C (5V POWER)
Page 25	25.Key_LED

radxa

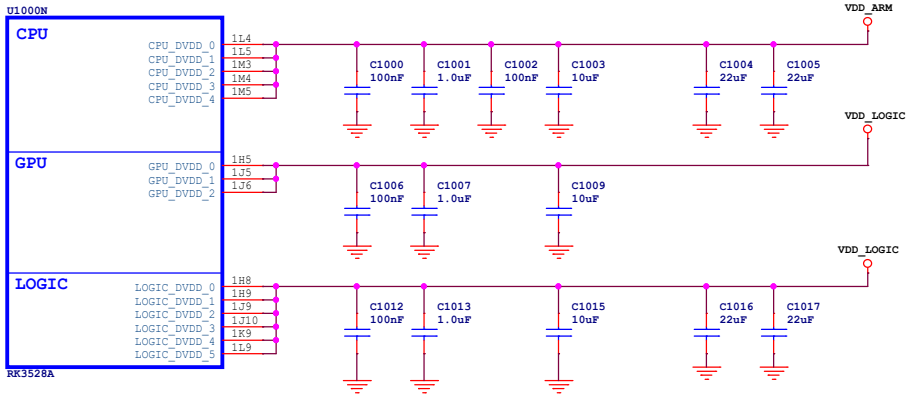
Size	Title:	REV
A4	ROCK 2F	V1.01
	Page Name: 02.Index	
Date: Monday, June 17, 2024	Sheet 02 of 25	

RK3528A Block Diagram

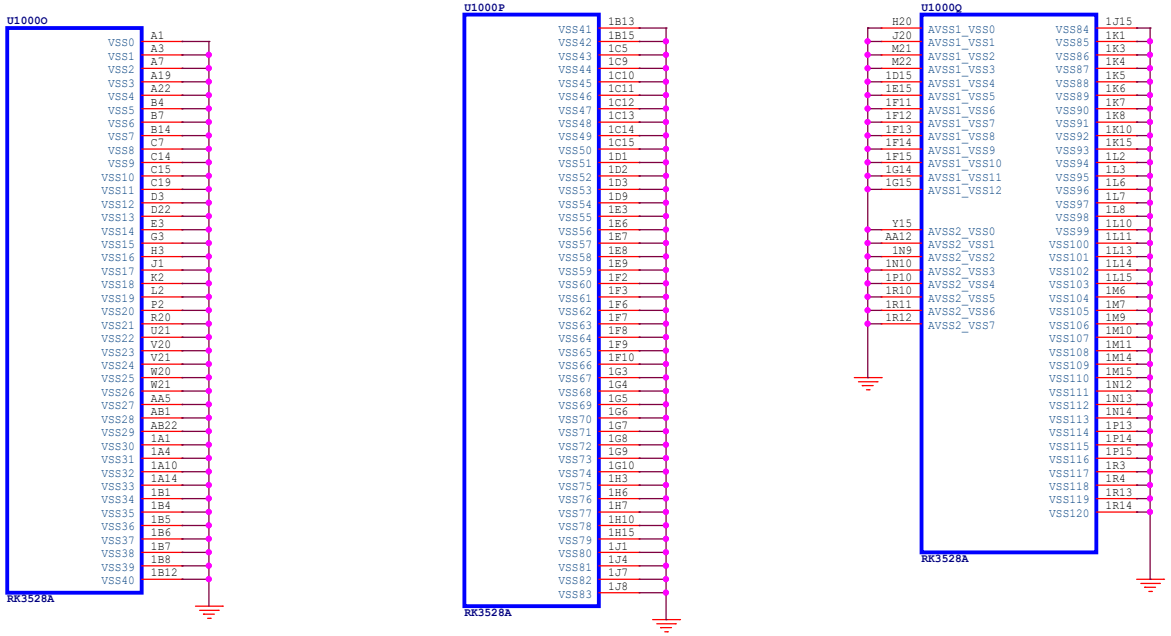


Size	Title:	ROCK 2F	REV
A2	Page Name:	03.Block	V1.01
Date:	Monday, June 17, 2024	Sheet 03 of 25	

RK3528A_N(POWER)

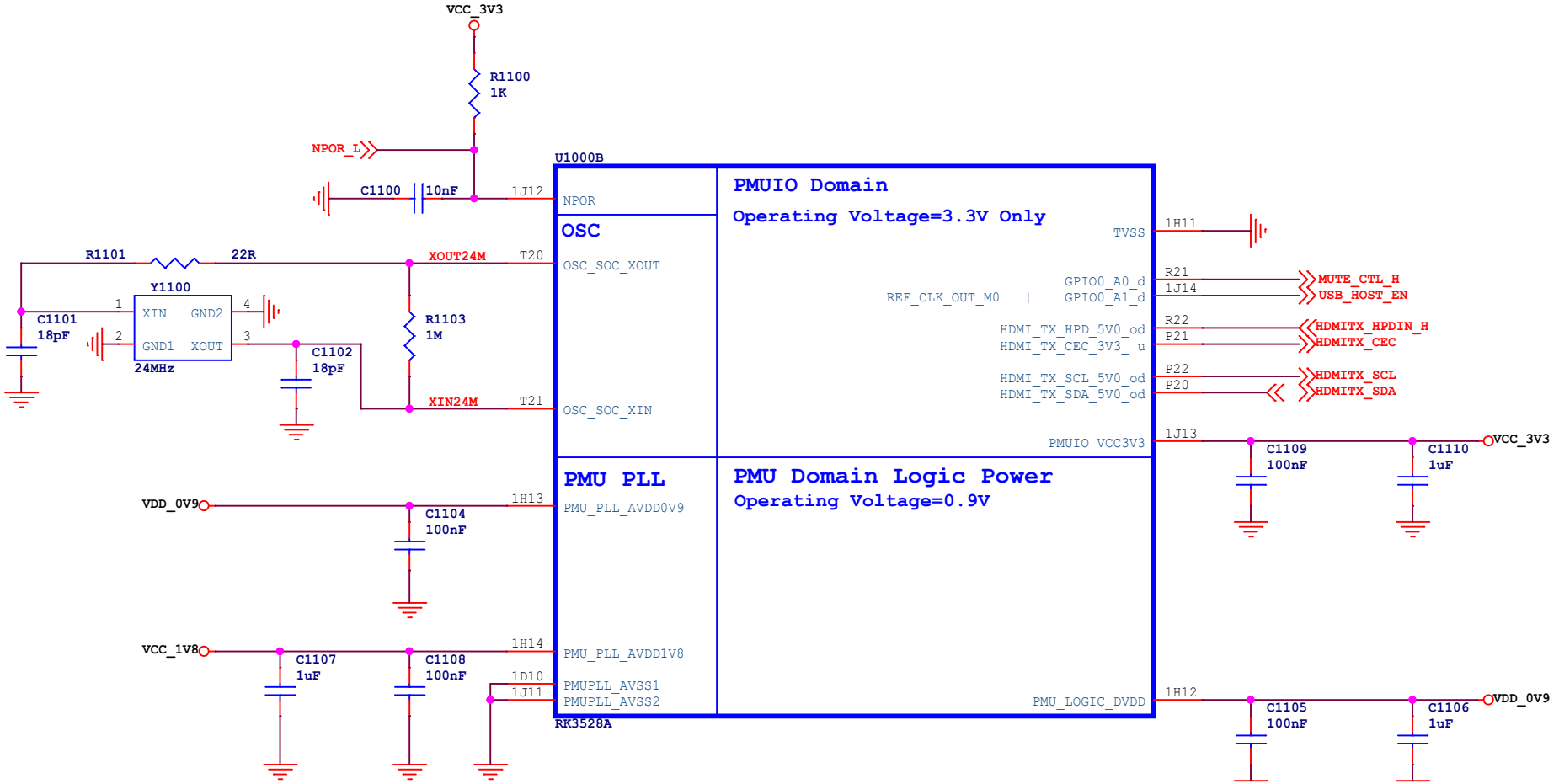


RK3528A_O/P/Q(GND)



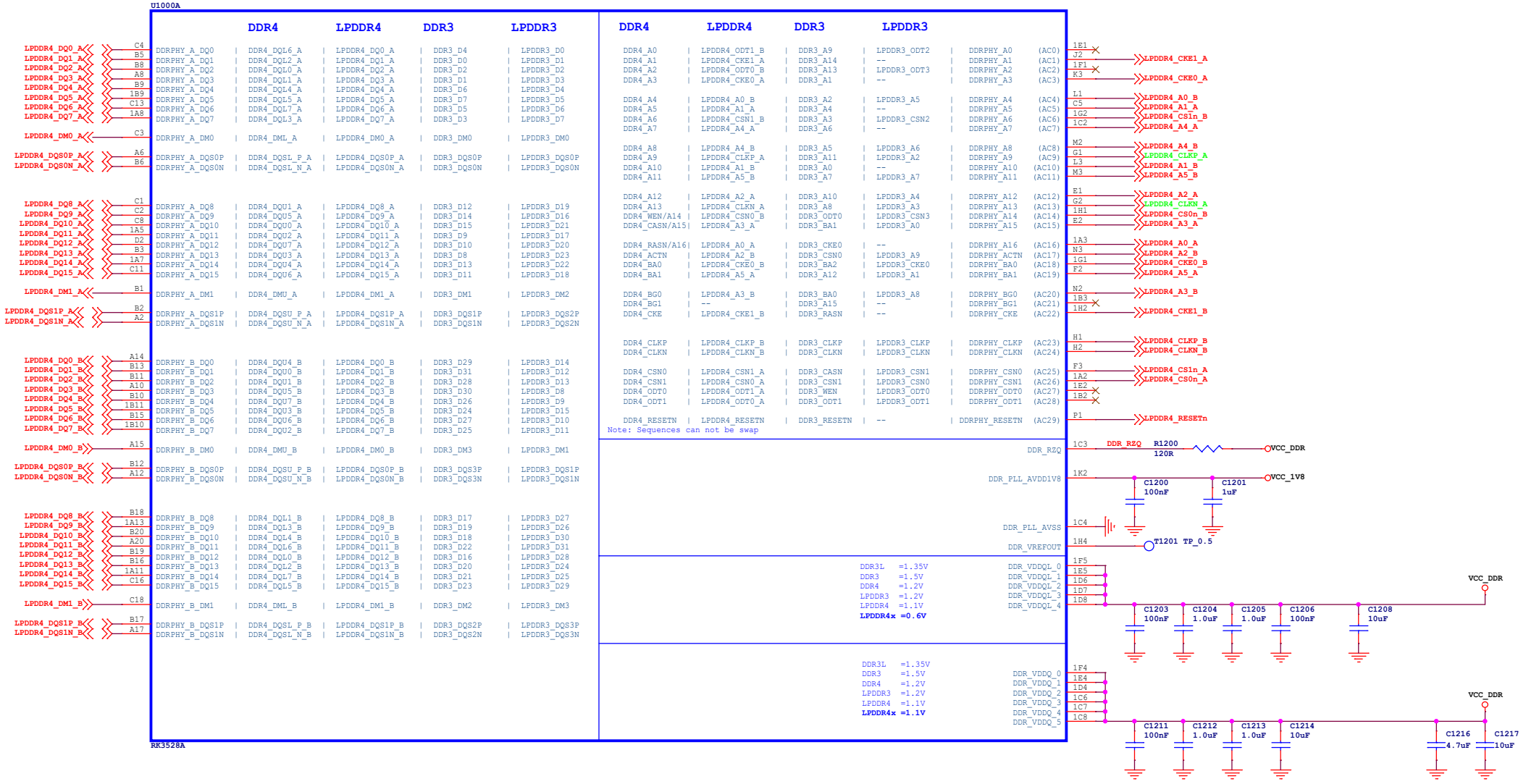
Size	Title: ROCK 2F		REV
A3	Page Name: 04.RK3528A Power/GND		V1.0
Date: Monday, June 17, 2024	Sheet 04 of 25		

RK3528A_B(OSC/PLL/PMUIO Domain)



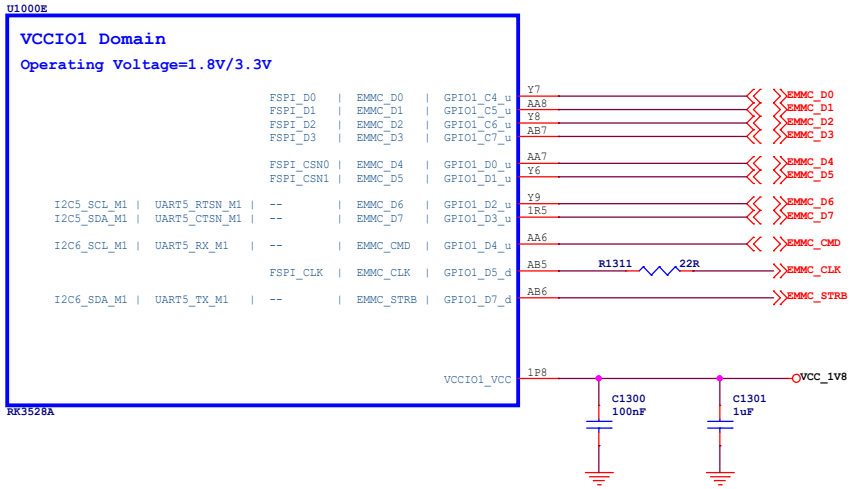
Size A4	Title: ROCK 2F	REV V1.01
	Page Name: 05.RK3528A OSC/PLL/PMUIO	
Date: Tuesday, June 18, 2024	Sheet 05 of 25	

RK3528A_A(DDR PHY)

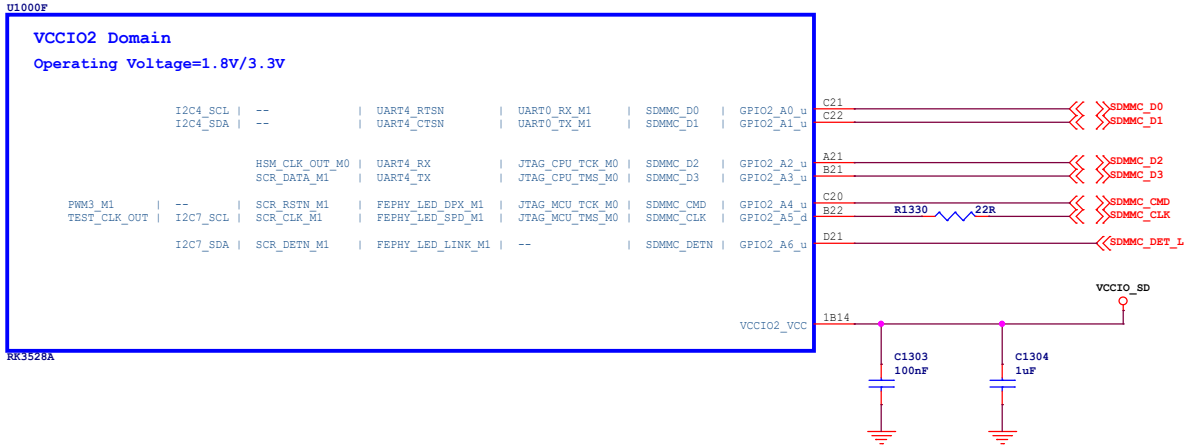


Size	Title: ROCK 2F		REV
A3	Page Name: 06.RK3528A DDR Controller		V1.0
Date: Monday, June 17, 2024	Sheet 06 of 25		

RK3528A_E(VCCIO1 Domain)

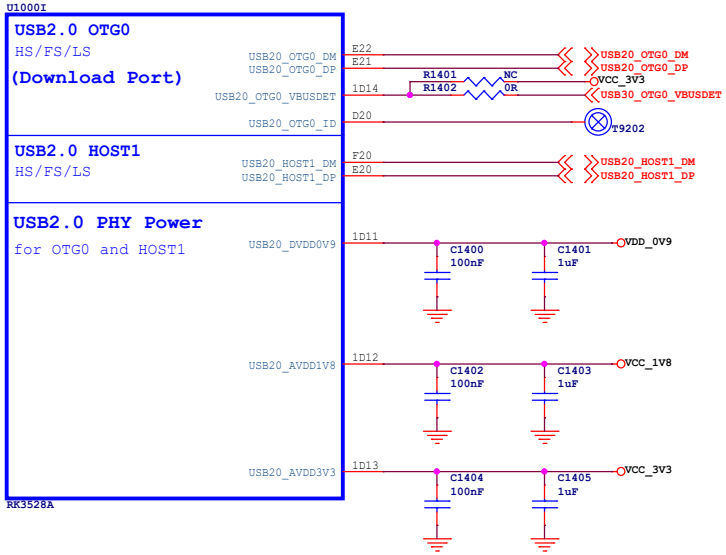


RK3528A_F(VCCIO2 Domain)

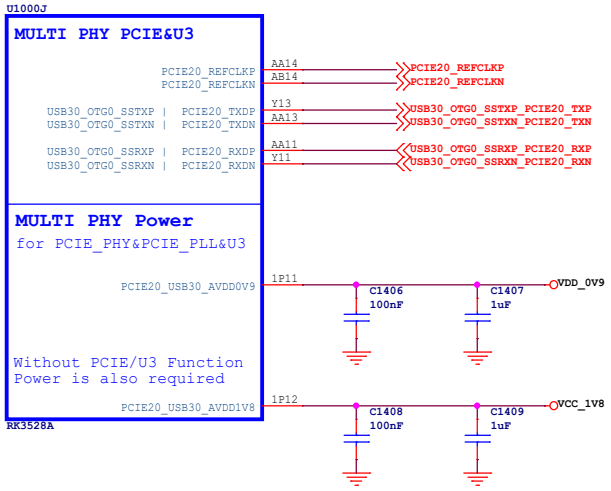


Size A3	Title: ROCK 2F	REV V1.0
	Page Name: 07.RK3528A FLASH/SD Controller	
Date: Monday, June 17, 2024	Sheet 07 of 25	

RK3528A_I(USB2.0 OTG/HOST)

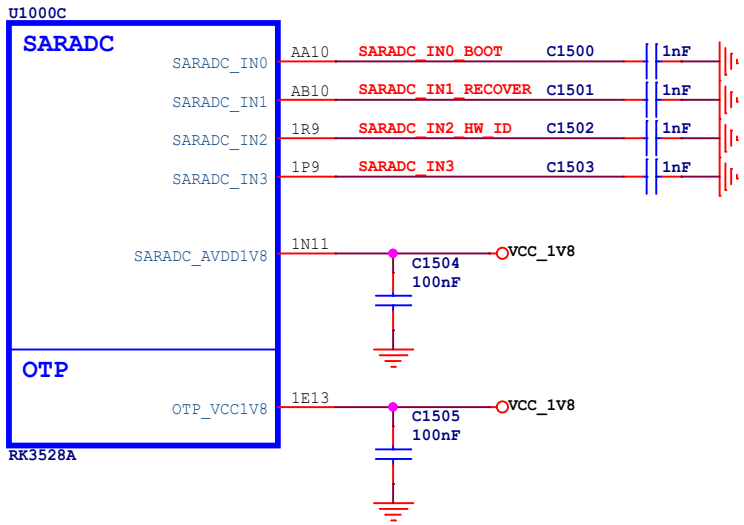


RK3528A_J(PCIE2.0/U3 PHY)



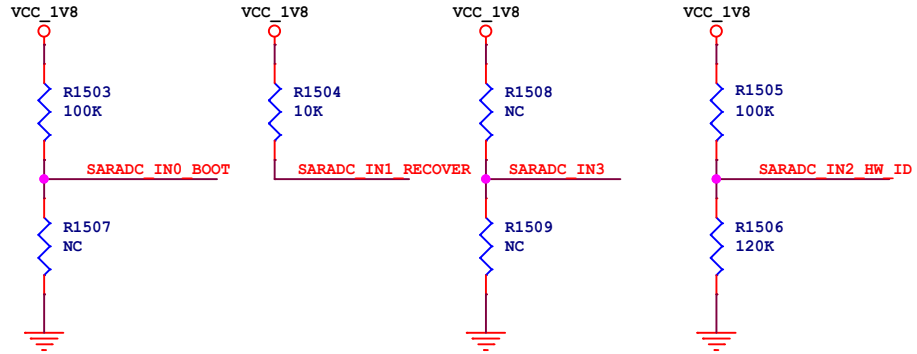
Size	Title: ROCK 2F	REV
A3	Page Name: 08.RK3528A USB/PCIE Controller	V1.0
Date: Monday, June 17, 2024	Sheet 08 of 25	

RK3528A_C(Saradc/OTP)



TABLE

Item	Rup	Rdown	ADC	BOOT MODE
LEVEL1	DNP	100K	0	USB (Maskrom mode)
LEVEL2	100K	12K	114	
LEVEL3	100K	27K	228	FSPI--USB
LEVEL4	100K	51K	342	
LEVEL5	100K	82K	456	
LEVEL6	100K	120K	570	EMMC--USB
LEVEL7	100K	200K	683	EMMC--SD Card--USB
LEVEL8	100K	330K	796	SD Card--USB
LEVEL9	100K	820K	910	
LEVEL10	100K	DNP	1023	FSPI--EMMC--SD Card--USB



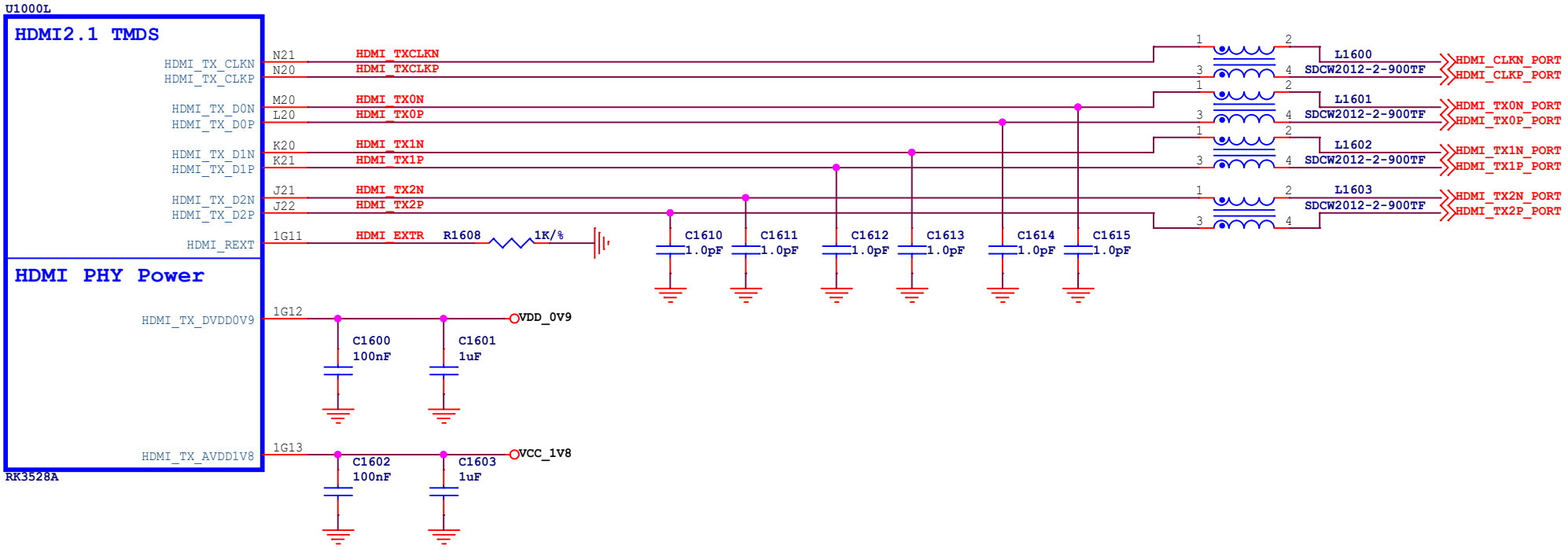
HW_ID	Rup	Rdown	Note
HW_ID1	100K	120K	onboard antenna
HW_ID2	100K	200K	external antenna



radxa

Size	Title:	REV
A4	ROCK 2F	V1.01
	Page Name: 09.RK3528A SARADC/OTP	
Date: Tuesday, June 18, 2024	Sheet 09 of 25	

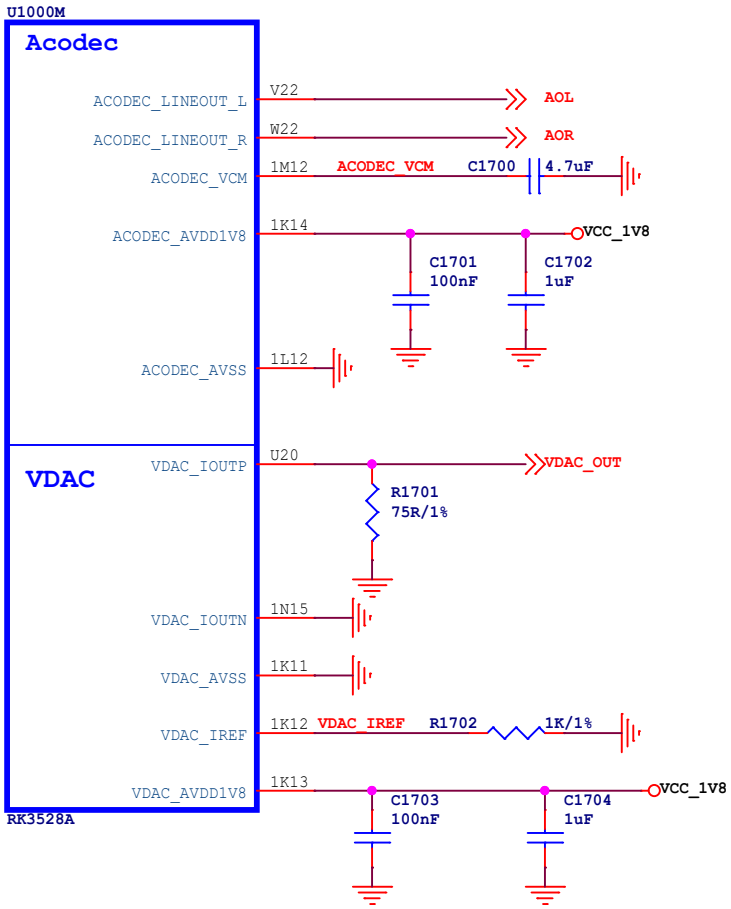
RK3528A_L(HDMI PHY)



radxa

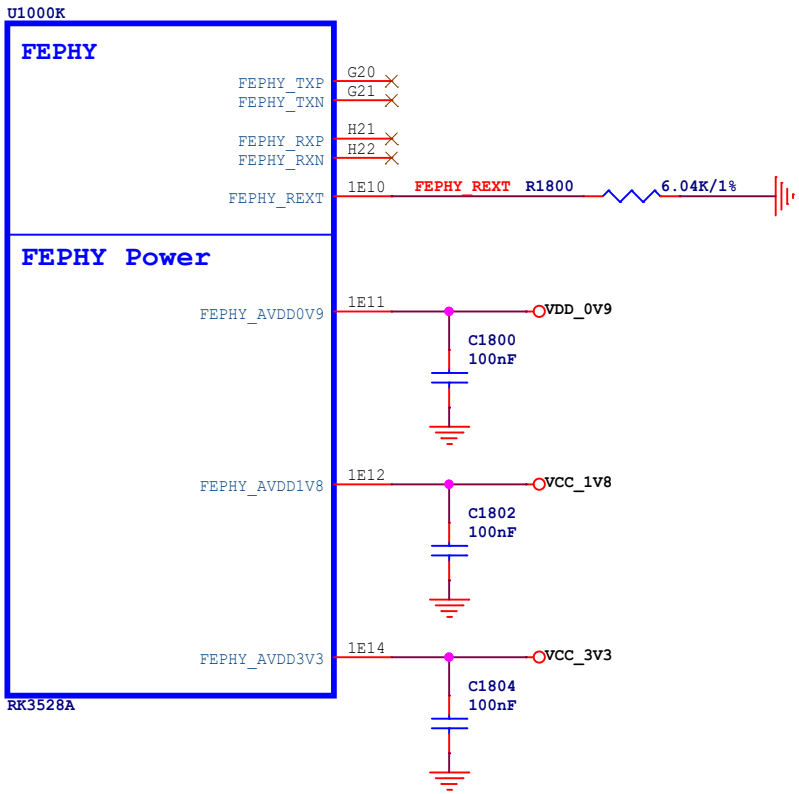
Size	Title:	REV
A4	ROCK 2F	V1.0
	Page Name: 10.RK3528A HDMI Interface	
Date: Monday, June 17, 2024	Sheet 10 of 25	

RK3528A_M(Acodec/VDAC)



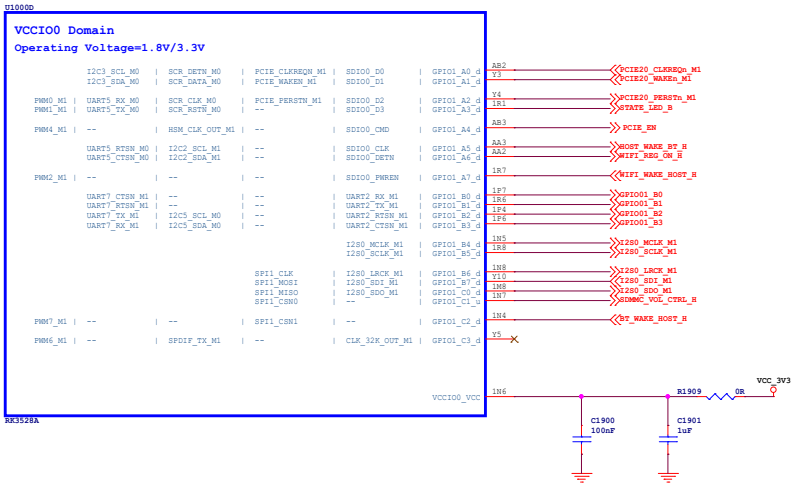
Size: A4	Title: ROCK 2F	REV: V1.01
	Page Name: 11.RK3528A AV OUT	
Date: Monday, June 17, 2024	Sheet 11 of 25	

RK3528A_K(Embed FEPHY)

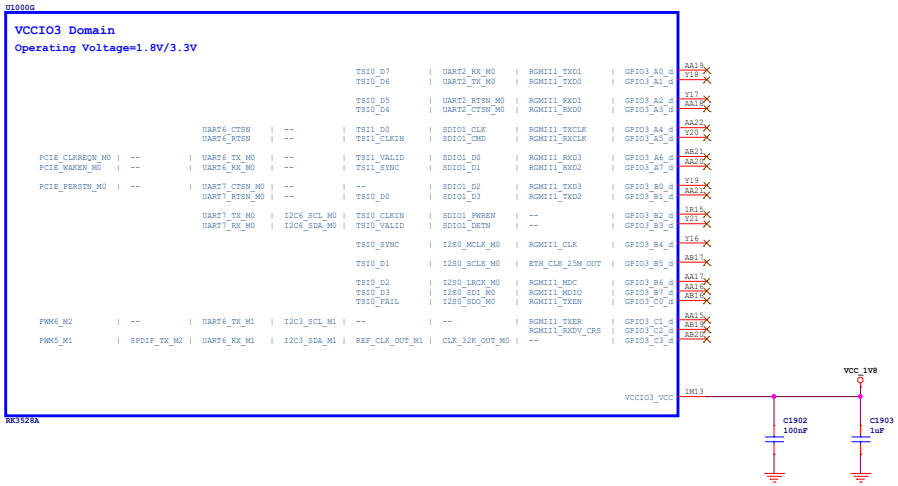


Size: A4	Title: ROCK 2F	REV: V1.01
	Page Name: 12.RK3528A Embed FEPHY	
Date: Monday, June 17, 2024	Sheet 12 of 25	

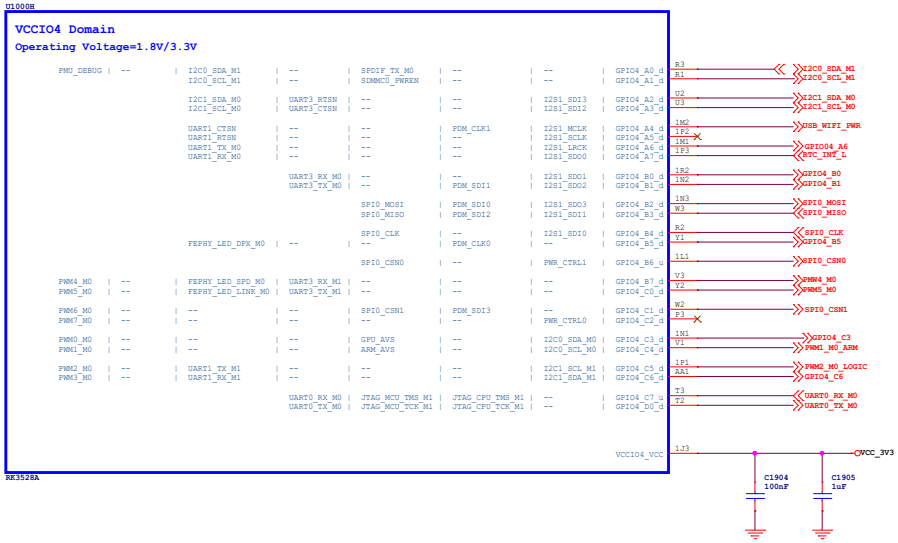
RK3528A_D(VCCIO0 Domain)



RK3528A_G(VCCIO3 Domain)

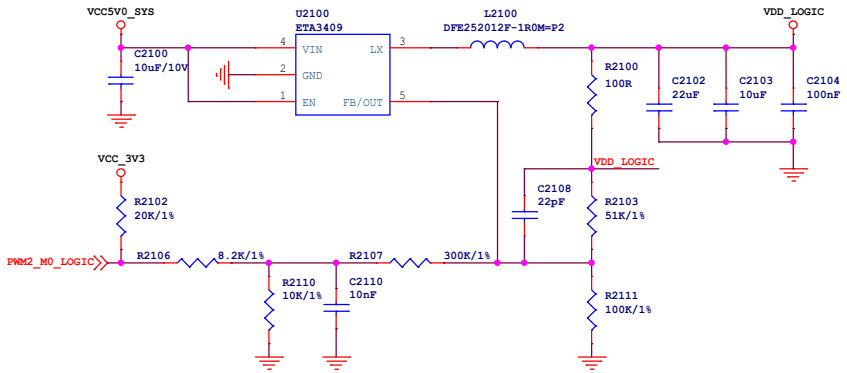


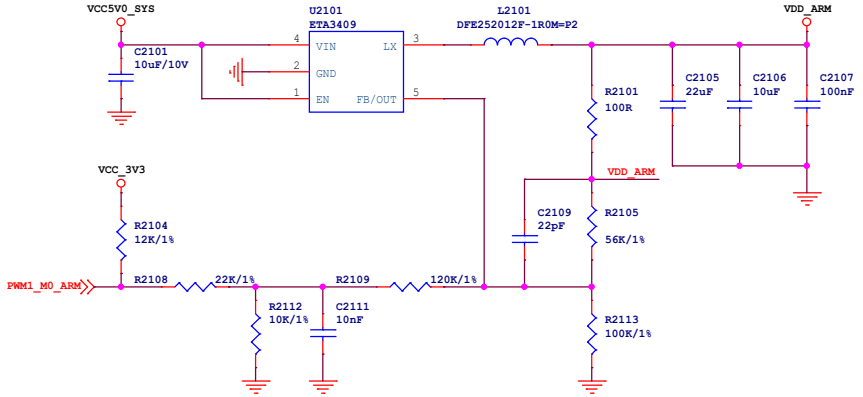
RK3528A_H(VCCIO4 Domain)

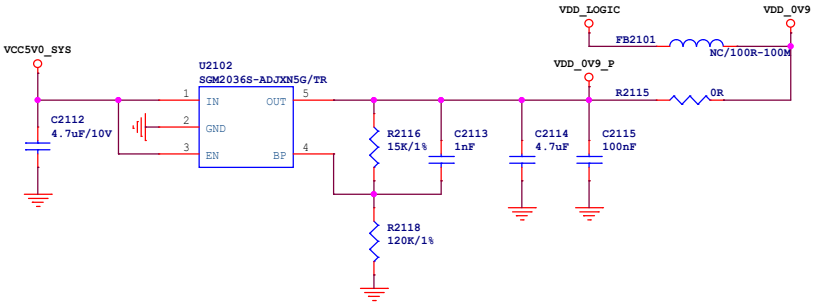


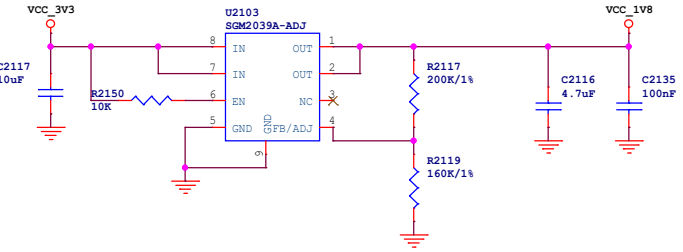
radxa

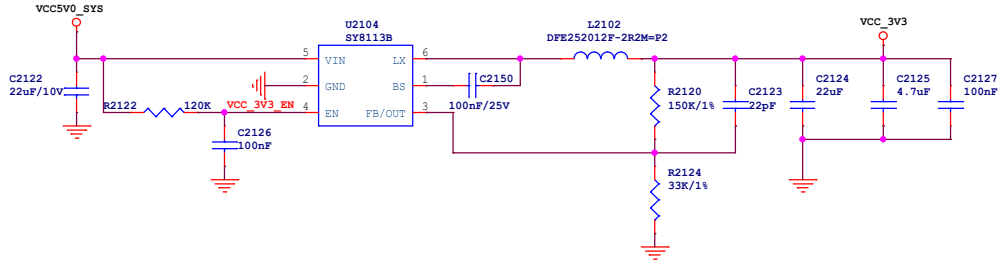
Size	Title:	ROCK 2F	REV
A2	Page Name:	13.RK3528A_1.8V/3.3V GPIO	V1.01
Date:	Monday, June 17, 2024	Sheet 13 of 25	

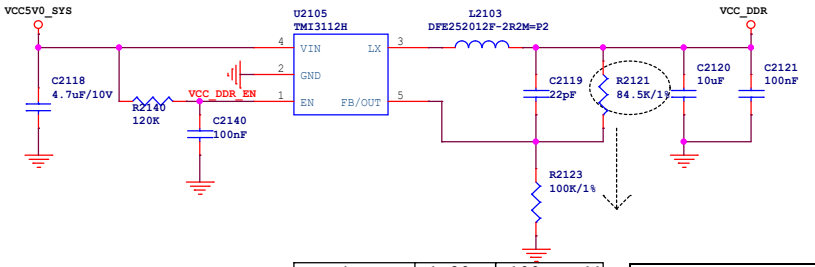










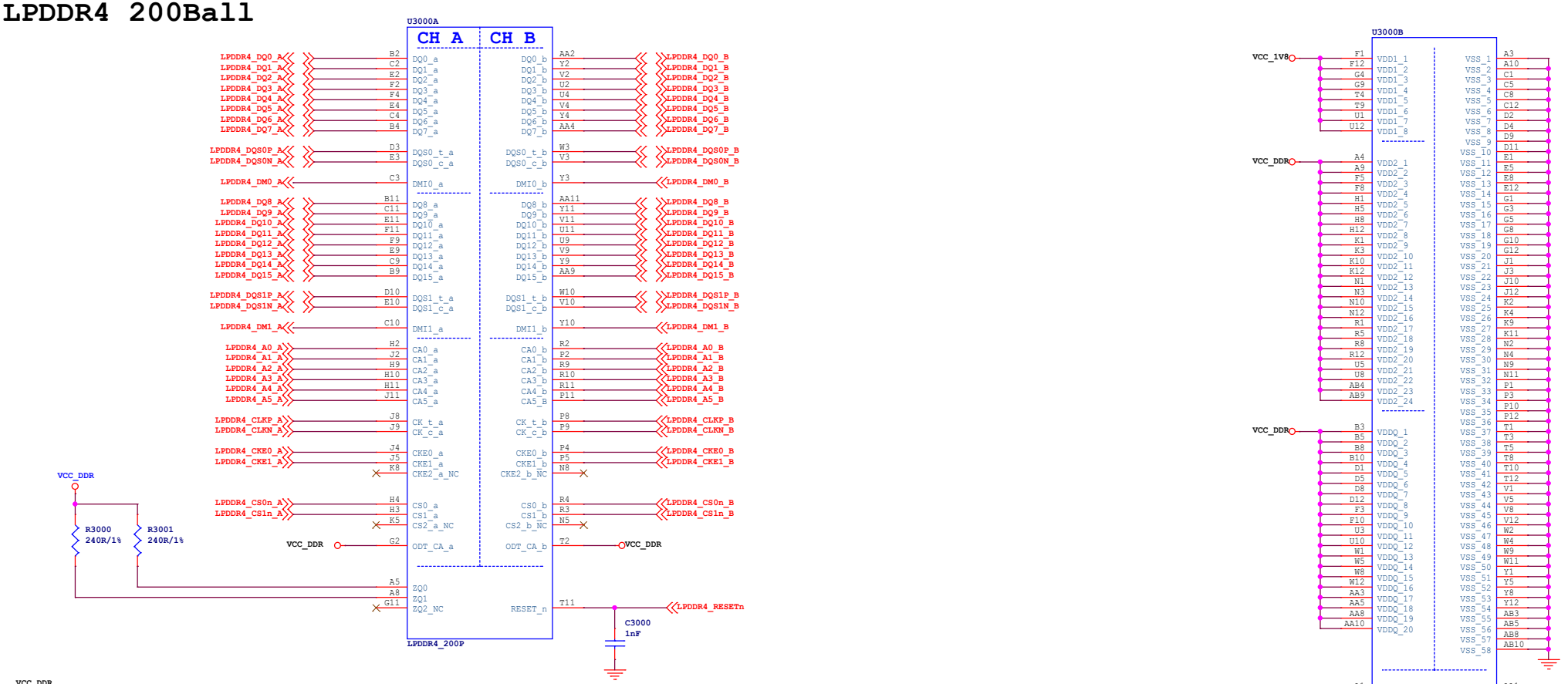


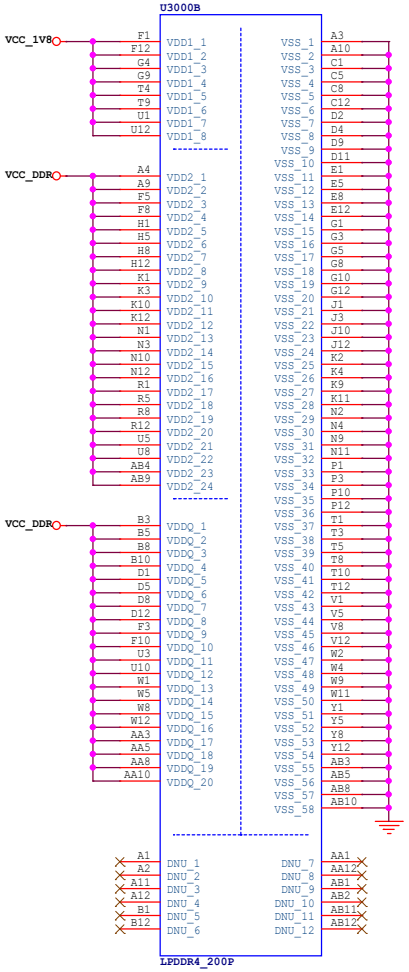
DDR4	1.20V	100K	1%
LPDDR4/4x	1.10V	84.5K	1%
DDR3	1.5V	150K	1%
DDR3L	1.35V	127K	1%
LPDDR3	1.25V	110K	1%

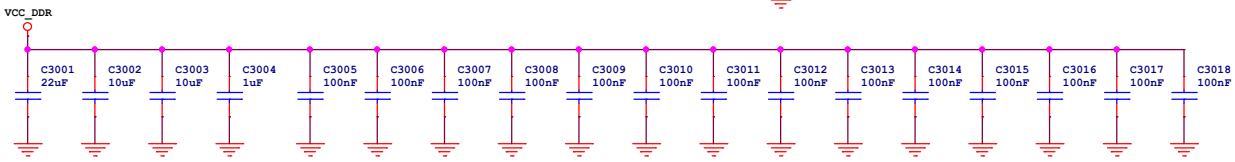
radxa

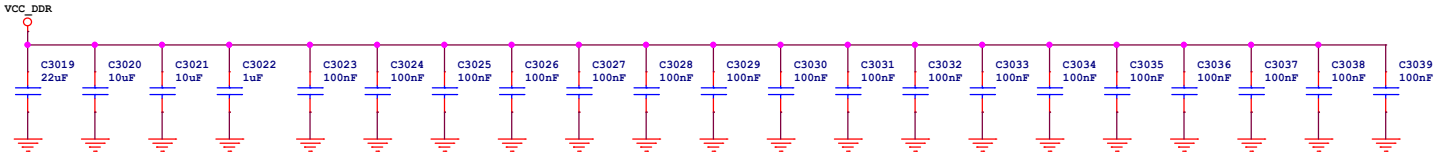
Size	Title: ROCK 2F	REV
A3	Page Name: 14.Power-DiscretePower	V1.0
Date: Tuesday, June 18, 2024	Sheet 14 of 25	

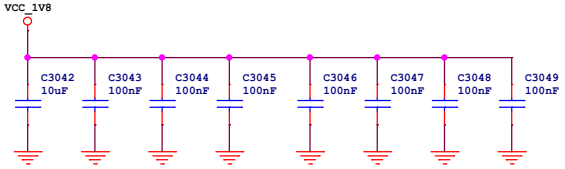
LPDDR4 200Ball





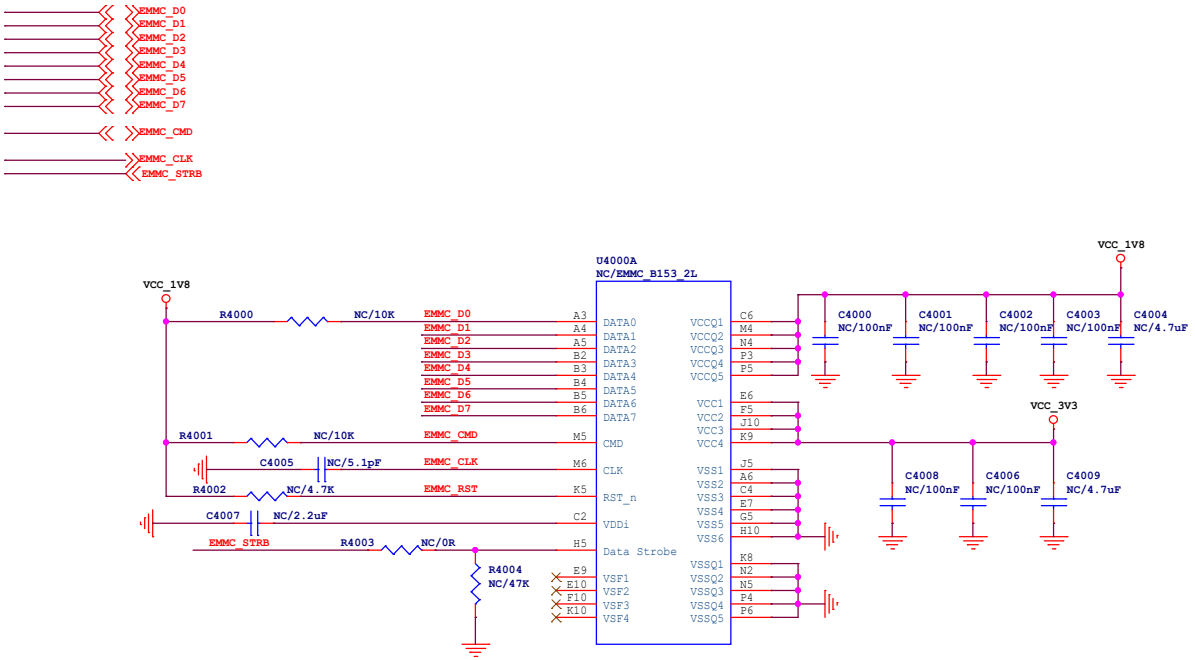


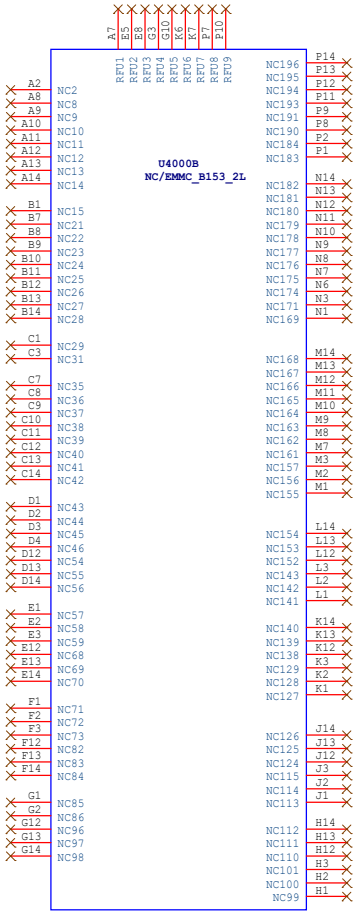




Size A3	Title: ROCK 2F	REV V1.0
	Page Name: 15.DRAM-LPDDR4_1X32bit_200P	
Date: Monday, June 17, 2024	Sheet 15 of 25	

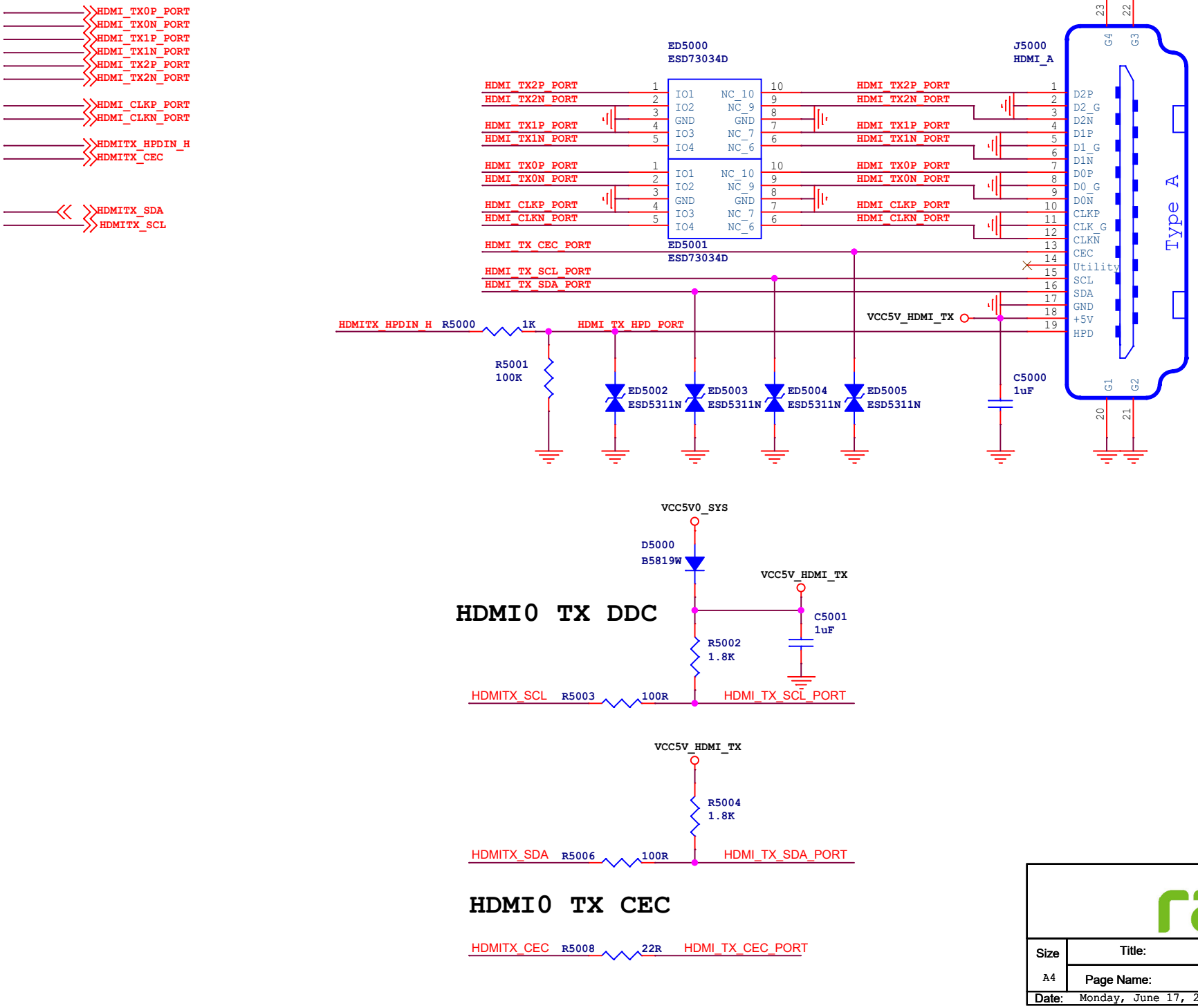
EMMC(option)





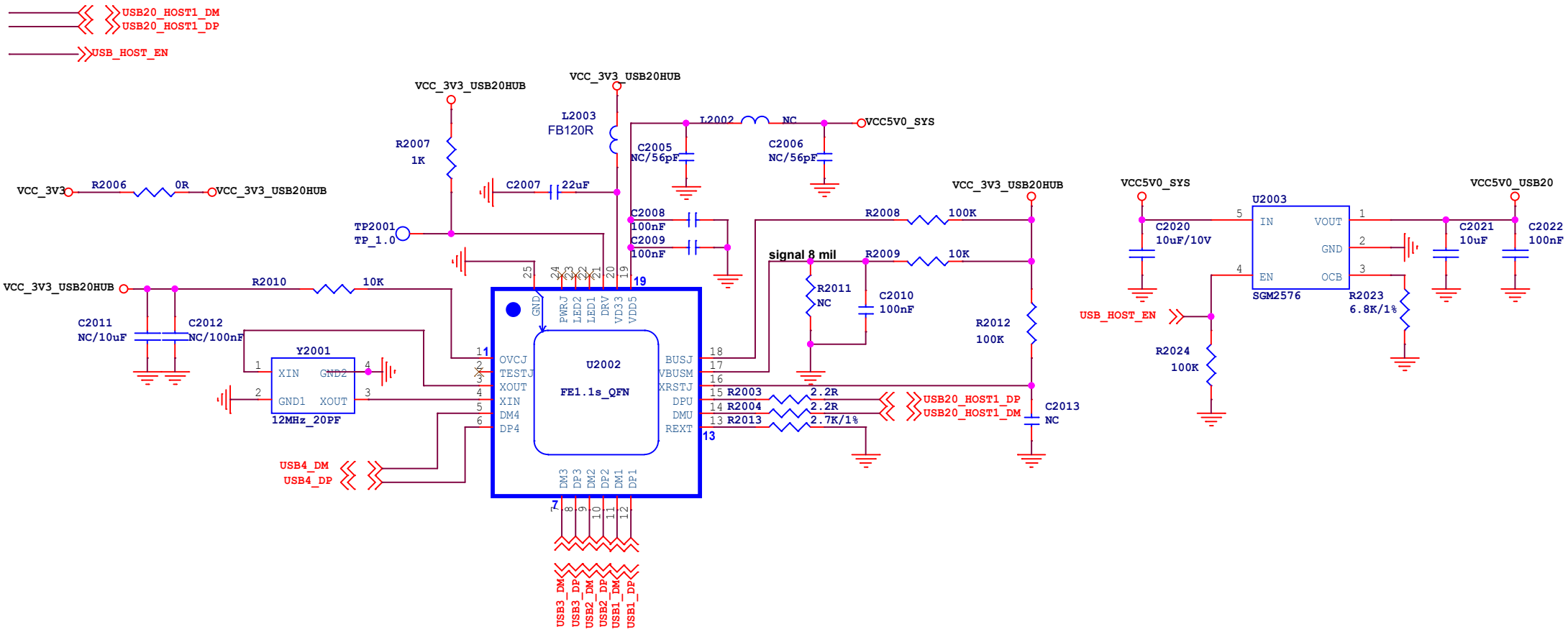
Size	Title: ROCK 2F		REV
A3	Page Name: 16.Flash-eMMC		V1.0
Date: Monday, June 17, 2024	Sheet 16 of 25		

HDMI2.0 TX

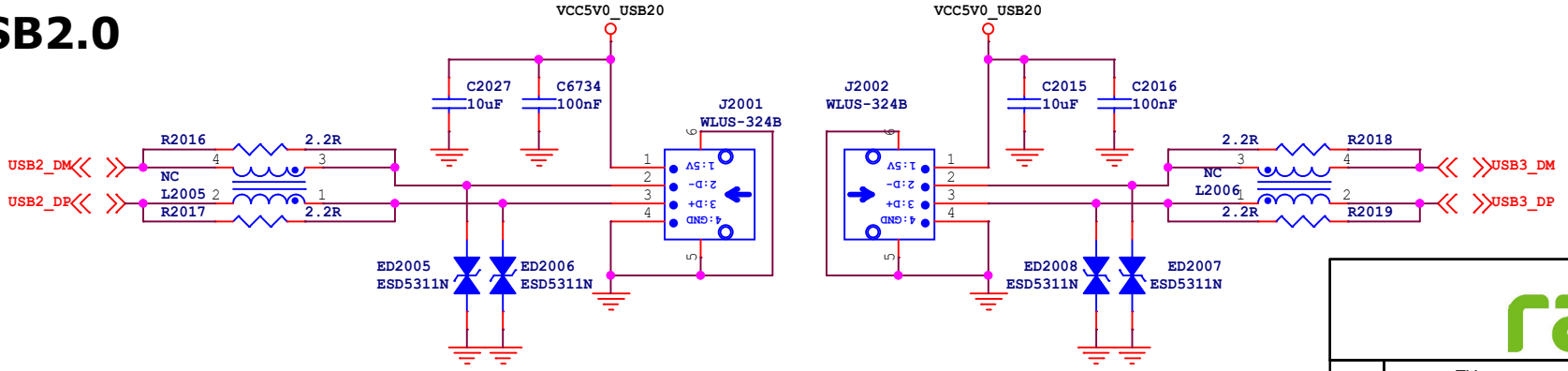


Size: A4	Title: ROCK 2F Page Name: 17.HDMI TX	REV: V1.01
Date: Monday, June 17, 2024	Sheet 17 of 25	

USB2.0 HUB

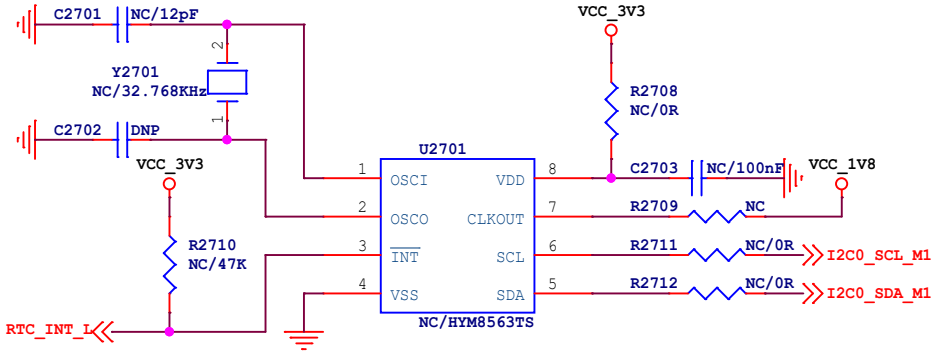


USB2.0

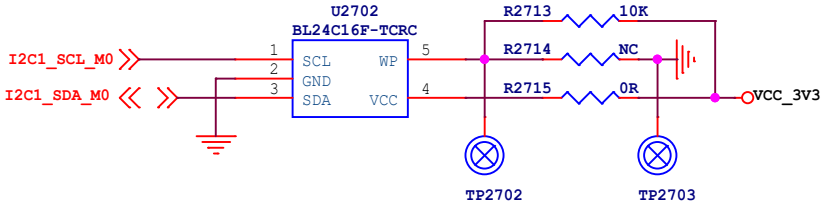


Size A4	Title: ROCK 2F	REV V1.01
	Page Name: 18.USB_HUB	
Date: Monday, June 17, 2024	Sheet 18 of 25	

RTC(option)

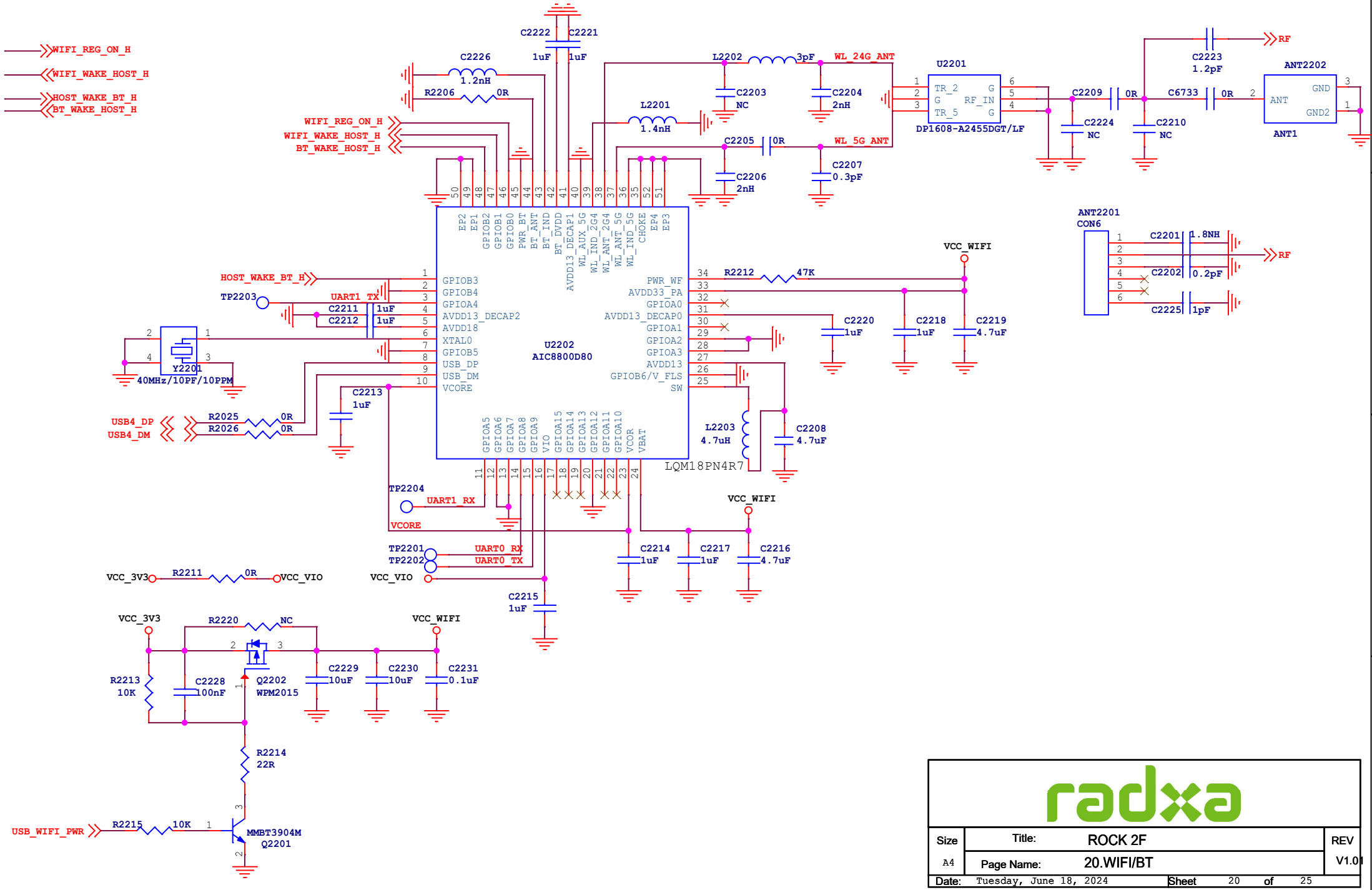


EEPROM



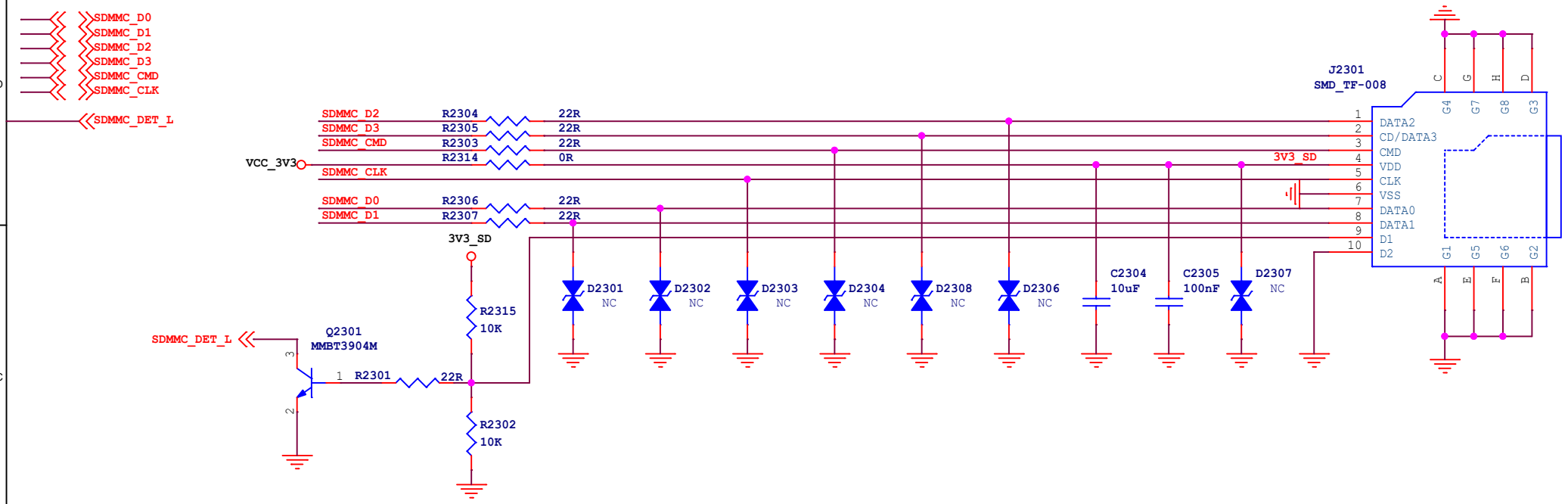
Size: A4	Title: ROCK 2F	REV: V1.01
	Page Name: 19.RTC_EEPROM	
Date: Monday, June 17, 2024	Sheet 19 of 25	

WIFI/BT

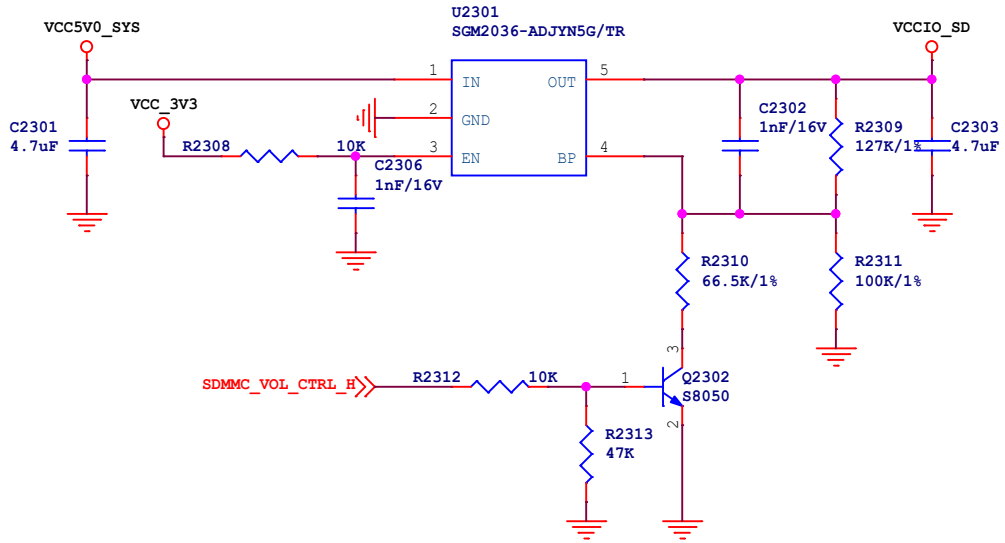


Size	Title:	ROCK 2F	REV
A4	Page Name:	20.WIFI/BT	V1.01
Date:	Tuesday, June 18, 2024	Sheet 20 of 25	

Micro_SD Card

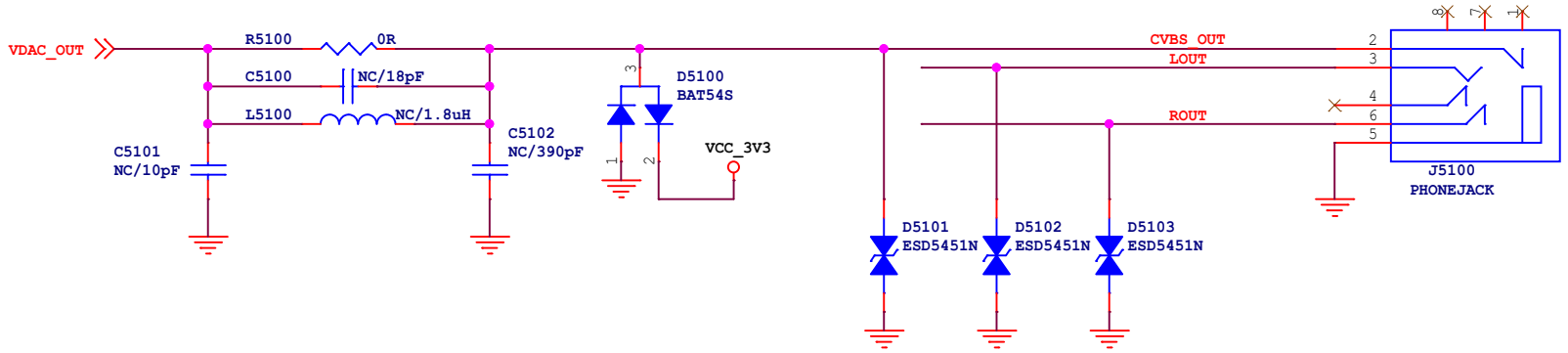


VCCIO_SD

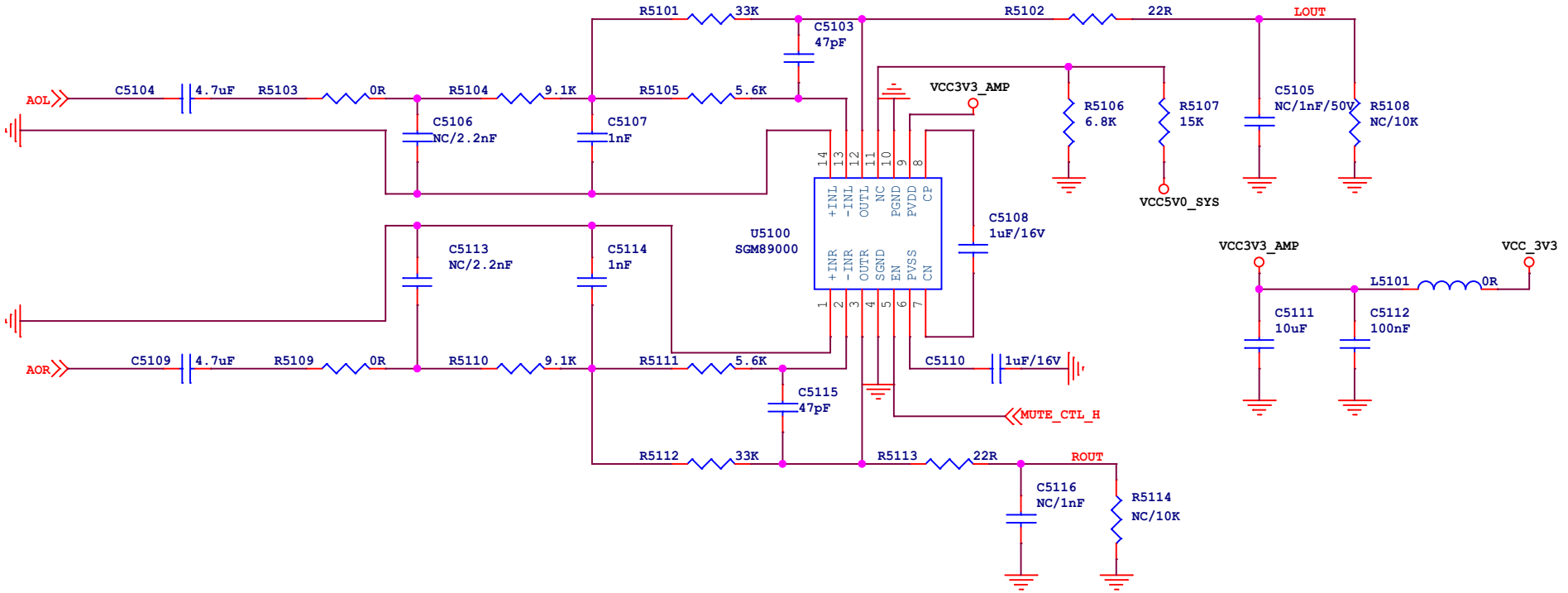


Size: A4	Title: ROCK 2F	REV: V1.01
	Page Name: 21.Micro_SD	
Date: Monday, June 17, 2024	Sheet 21 of 25	

CVBS/AV_OUT

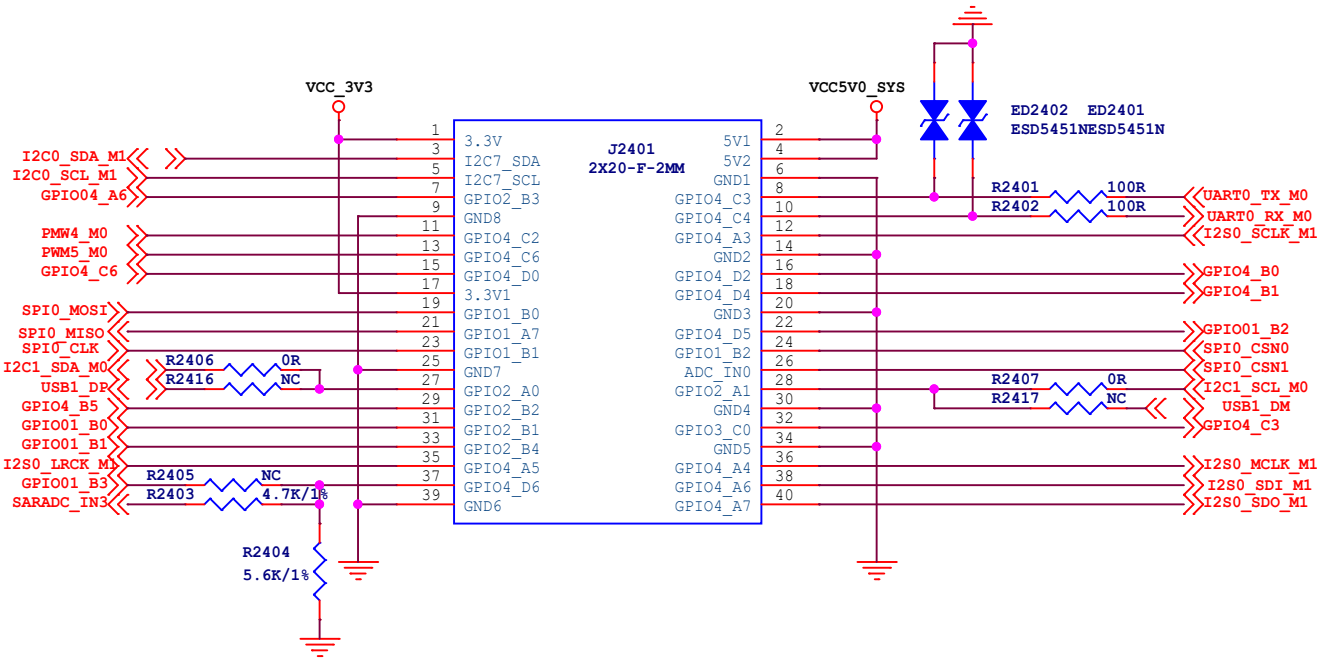


2-Vrms Audio Line Driver

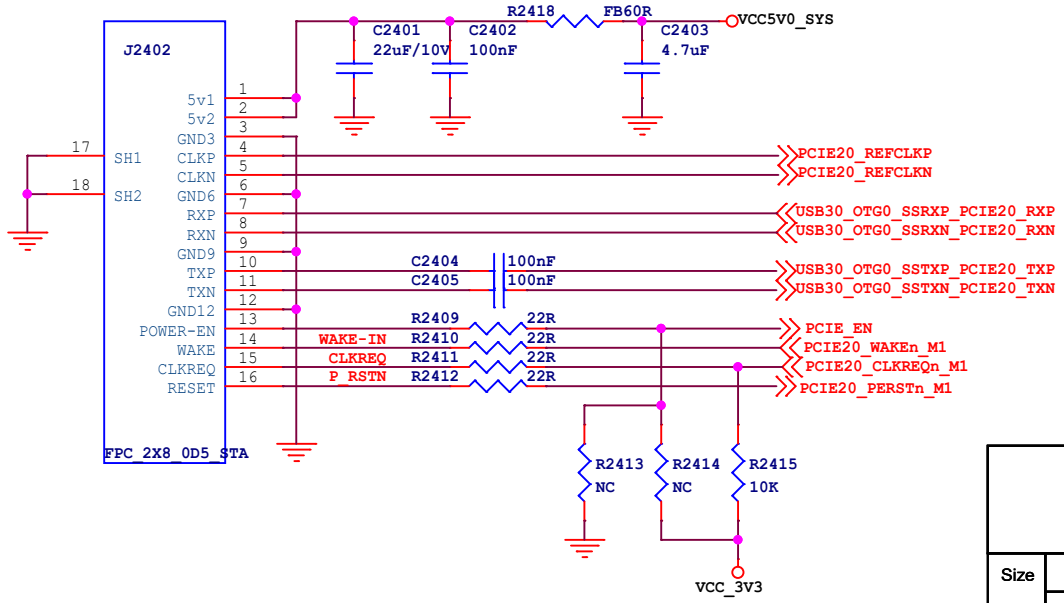


Size	Title:	ROCK 2F	REV
A4	Page Name:	22.VO-AV Interface	V1.01
Date:	Monday, June 17, 2024	Sheet 22 of 25	

40PIN_Connctor

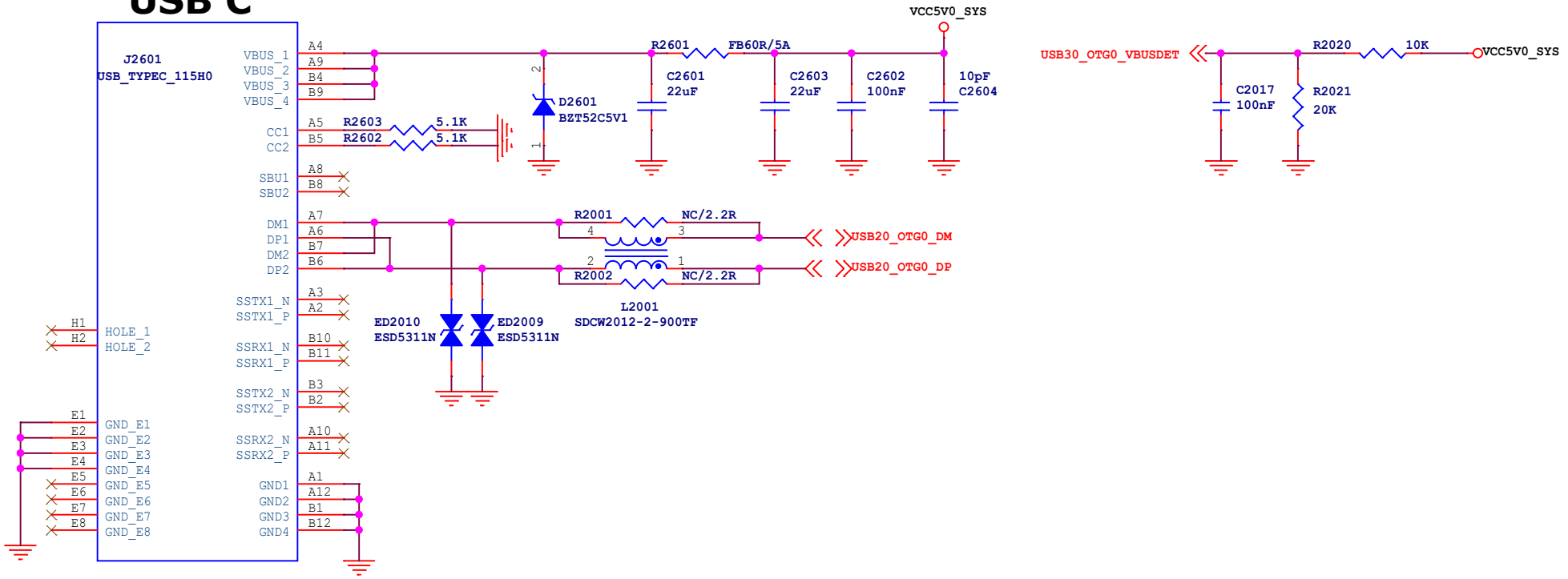


PCIE interface



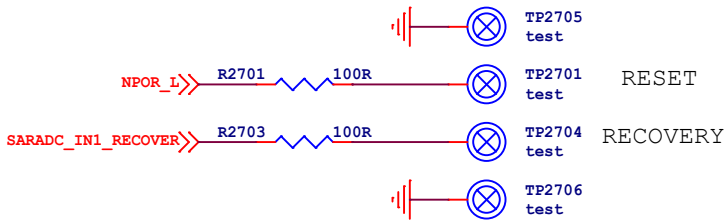
Size A4	Title: ROCK 2F	REV V1.01
	Page Name: 23.40PIN_Connctor/PCIE interface	
Date: Monday, June 17, 2024	Sheet 23 of 25	

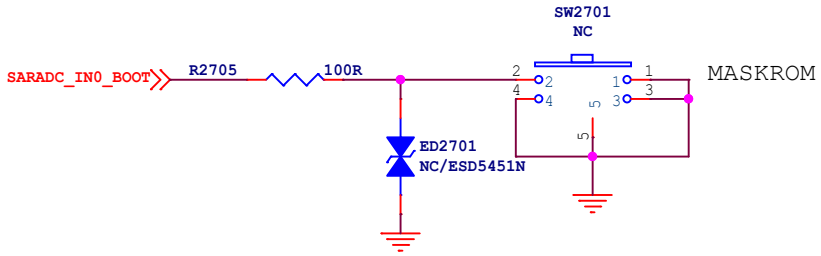
USB C



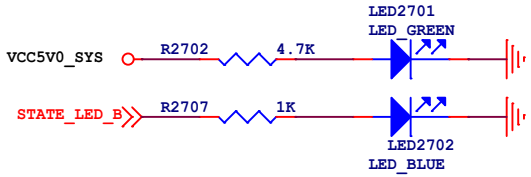
Size: A4	Title: ROCK 2F	REV: V1.01
	Page Name: 24.USB C (5V POWER)	
Date: Monday, June 17, 2024	Sheet 24 of 25	

KEY





LED



Size: A4	Title: ROCK 2F	REV: V1.01
	Page Name: 25.Key_LED	
Date: Tuesday, June 18, 2024	Sheet 25 of 25	